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import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Input

import kagglehub
import os

# Download dataset
path = kagglehub.dataset_download("vikrishnan/iris-dataset")
print("Path to dataset files:", path)

# 🔍 Check actual file name (VERY IMPORTANT)
files = os.listdir(path)
print("Files in dataset:", files)

# Load dataset (automatically picks correct file)
df = pd.read_csv(os.path.join(path, files[0]))

# Preview data
print(df.head())

# Features and labels
X = df.iloc[:, :-1].values
y = df.iloc[:, -1].values

# Encode labels (setosa, versicolor, virginica → 0,1,2)
le = LabelEncoder()
y = le.fit_transform(y)

# Normalize features
scaler = StandardScaler()
X = scaler.fit_transform(X)

# Train-test split (fixed)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Build model
model = Sequential([
    Input(shape=(4,)),
    Dense(16, activation='relu'),
    Dense(16, activation='relu'),
    Dense(3, activation='softmax')
])

# Compile model
model.compile(
    optimizer='adam'
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optimizer = optim.Adam ,
loss='sparse_categorical_crossentropy',
metrics=['accuracy']
)

# Summary
model.summary()

# Train model
model.fit(
    X_train, y_train,
    epochs=50,
    batch_size=8,
    validation_split=0.1,
    verbose=1
)

# Evaluate model
test_loss, test_acc = model.evaluate(X_test, y_test)

print("Test accuracy:", test_acc)
```

